

Photosynthesis: Capturing Light Energy

- Photosynthesis converts light energy into chemical energy stored in glucose.
- It occurs mainly in chloroplasts. Chlorophyll in the thylakoid membranes absorbs light.
- The light-dependent reactions split water, release oxygen, and produce ATP and NADPH.
- These energy carriers power the Calvin cycle in the chloroplast stroma.
- Key equation: carbon dioxide + water + light $\rightarrow$ glucose + oxygen.

Cellular Respiration: Releasing Stored Energy

- Cellular respiration breaks down glucose to make ATP that cells can use.
- Glycolysis begins in the cytoplasm and turns one glucose into two pyruvate molecules.
- The citric acid cycle and electron transport chain continue inside mitochondria.
- Oxygen is the final electron acceptor in aerobic respiration, forming water.
- Photosynthesis stores energy; respiration releases it. Their products and reactants connect the two processes.

How Plants Balance Both Processes

- Plants perform both photosynthesis and cellular respiration.
- In daylight, photosynthesis can produce glucose and oxygen faster than respiration consumes them.
- At night, photosynthesis stops without light, but cellular respiration continues.
- Stomata regulate gas exchange: carbon dioxide enters and oxygen and water vapor can leave.
- Together, these pathways cycle matter while energy flows from sunlight to ATP-powered cell work.